

OpenEarable 1.4: Dual Microphones Earpiece to Capture In-Ear and Outer-Ear Audio Signals

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ABSTRACT

In recent years, earables have emerged as a focal point for exploring novel sensing principles on the ears. OpenEarable, an open reference platform, has been developed to integrate various sensors in and around the ears. However, it has thus far lacked the capability to capture audio both inside the ear canal and outside simultaneously. To address this gap, we introduce a novel earpiece for OpenEarable equipped with two microphones, designed to mimic commercial earbuds like AirPods, which commonly restrict access to internal microphones. This new earpiece utilizes the interchangeable 8-pin connector earpiece concept of OpenEarable, enabling the exploration of algorithms that leverage both internal and external sounds captured at the ears (e.g., noise canceling). By updating the existing app and dashboard, OpenEarable 1.4 seamlessly integrates with the existing suite of devices within the OpenEarable ecosystem. We validate the earpiece by showcasing its abilities to occlude the ear canal and measure the heartbeat of a wearer. The enhancement of OpenEarable facilitates streamlined and large-scale audio data collection, advancing the development of novel earable technologies.

CCS CONCEPTS

• **Hardware** → **Emerging technologies**; • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **Ubiquitous and mobile devices**;

KEYWORDS

open-source; earables; hearables; open-source hardware; OSHW

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1 INTRODUCTION AND RELATED WORK

In recent years, the field of earables has seen significant advancements, primarily driven by integration of novel sensing principles such as IMUs [17], PPG sensors [8], temperature sensors [20], in-ear [5] and outer ear [21] microphones, and bioelectric potential sensing [10]. These sensors make use of the unique anatomical properties of the ears to enable a large variety of applications [14].

One platform that has contributed to the growth of the earables field is OpenEarable [16], an open reference platform designed to integrate different sensors in and around the ears: a 9-axis IMU, an in-ear pressure sensor, and an in-ear ultrasound microphone and speaker. Despite its versatility, OpenEarable 1.3 has, until now, been limited by its inability to capture audio both inside the ear canal and from the surrounding environment. This limitation has restricted researchers' ability to develop and test algorithms that require audio data, similar to those used for, e.g. noise canceling, in commercial earbuds like AirPods Pro¹.

To address this gap, we introduce a novel earpiece for OpenEarable, equipped with dual microphones capable of capturing both internal and external sounds. This new earpiece leverages the 8-pin connector of the OpenEarable platform, thus enabling advanced research into auditory processing and interaction algorithms that can utilize a richer set of audio data. By updating the accompanying app and dashboard, OpenEarable 1.4 integrates seamlessly with the existing ecosystem, allowing for more efficient and large-scale data earable collection.

In sum, in this paper we contribute to open wearables and advance the OpenEarable platform by the:

- Introduction of a novel earpiece to OpenEarable that can sense both internal and external sounds using two ultrasound-capable microphones while still including an in-ear pressure sensor and a speaker.
- Validation of the occlusion effect of the earpiece by showing its potential to measure low frequency sounds such as heart sounds inside the ear canal.

¹<https://www.apple.com/de/airpods-pro/>

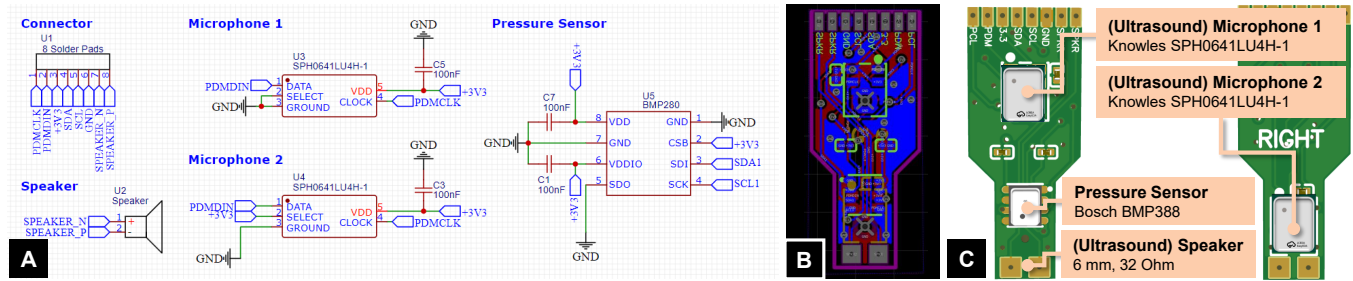


Figure 1: (A) Schematic diagram of the OpenEarable 1.4 dual microphone earpiece, showing the connections for microphones, speaker, and pressure sensor; (B) PCB layout of the right earpiece; (C) Assembled right earpiece with labeled components.

2 DESIGN PRINCIPLES

We extend OpenEarable 1.3 by redesigning the earpiece to include two microphones (one inward, one outward) while keeping the ear canal pressure sensor and improving the speaker for better bass response. We adhere to the best practices of the Open-Source Hardware Association (OSHA) [13] by (i) sharing all original hardware design source files and ready-to-view outputs (e.g. PDFs); (ii) using freely available tools, common production processes and widely available components for all designs; (iii) open-sourcing all firm- and software; (iv) providing photos and instructions how to make and use the hardware; (v) licensing all resources under MIT license². Components are selected based on proven parts from related hardware which are known to have good availability, primarily based on OpenEarable 1.3 [16].

3 HARD- AND SOFTWARE

The following sections describe the hard- and software, and the mechanical assembly of OpenEarable 1.4 dual microphone earpiece.

3.1 Hardware

The following section first introduces the OpenEarable foundation for which the earpiece presented in this paper was designed. We then describe the newly design earpiece and its assembly.

3.1.1 OpenEarable 1.3. The earpiece introduced in this paper extends the existing OpenEarable 1.3³ behind-the-ear main hardware which contains the MCU (nrf52840), a 9-axis IMU (Bosch BMX160), a microSD card slot, a battery and the charging controller. The OpenEarable 1.3 behind-the-ear part comes with an 8-pin JST 08SR-3S connector making it possible to swap the earpiece to connect to I2C, audio DAC, PDM (for microphones) and 3.3 V power supply.

3.1.2 Earpiece. The main hardware contribution of this paper is a novel earpiece. It integrates two Knowles SPH0641LU4H-1 microphones. The microphones are connected via PDM. One microphone is configured to be sampled on the rising edge of the PDM clock and one on the falling edge. This makes it possible to share the data line between the two microphones. The audio signal is recorded to the microSD card inside OpenEarable at up to 16,000 kHz per each microphone if recording simultaneously. Individually, each

microphone can be sampled at up to 62,500 kHz. The earpiece also contains the Bosch BMP280 pressure sensor which can measure ear-canal shape deformations based on pressure changes. In addition, the earpiece also contains a speaker (6 mm diameter) that is commonly available from AliExpress. It has a rated power of 3 mW, rated impedance of $32 \Omega \pm 15\%$ and sensitivity of 92 ± 3 dB. It is designed to operate in a frequency range of 20 Hz - 20 KHz but can also produce higher frequencies. Because of its enclosure, it can produce bass better than the OpenEarable 1.3 earpiece.

3.1.3 Mechanical Design and Assembly. Figure 2 illustrates the assembly of the OpenEarable 1.4 earpiece. The design consists of a front and back piece that encase the PCB, creating two separate chambers: one for the in-ear microphone and one for the outward-facing microphone. This configuration ensures optimal separation and functionality of the microphones. The enclosure can be 3D printed using a standard Formlabs 3B SLA printer. To achieve an airtight seal, the earpiece components are bonded with Pattex Ultra Gel Glue. We open-source the STL and source files for the enclosure.

3.2 Software

We describe the firmware running on OpenEarable and the web dashboard and app that connects to the device.

3.2.1 Firmware. The firmware of OpenEarable is based on Arduino [16]. This firmware was adapted to OpenEarable 1.4 to handle two microphones via PDM. Via BLE, it is possible to configure the sampling rates of each microphone independently. The updated firmware based on the 1.4 earpiece advertises as version 1.4 which can be detected by clients that connect to the earable. To make it easier to deploy the firmware compared to the OpenEarable 1.3 version, which required a series of manual changes in different libraries, we have ported the firmware to a PlatformIO⁴ project containing all necessary files. Additionally, firmware changes to version 1.4 enables writing of all sensor data, including the IMU and pressure sensor, to the SD card and changing of the microphones gain via BLE, both to enable efficient data collection for research purposes.

²<https://github.com/OpenEarable/open-earable>

³<https://open-earable.teco.edu>

⁴<https://platformio.org>

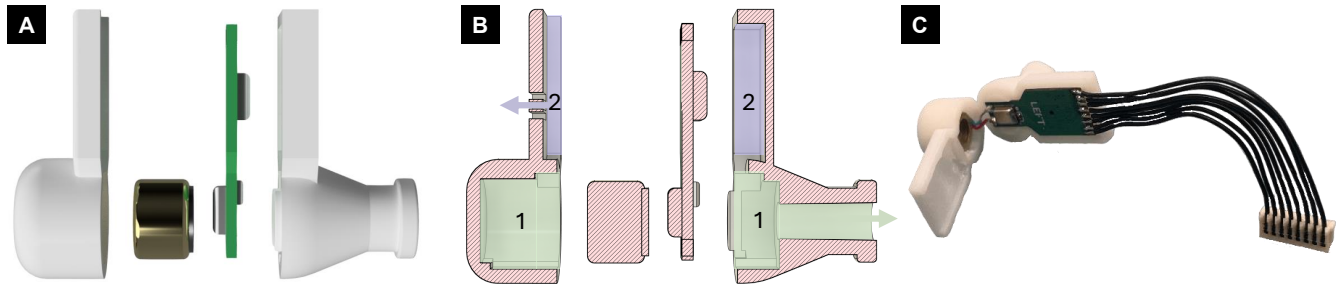


Figure 2: (A) Exploding rendering of the dual microphone earpiece; (B) a cross section view of the earpiece, revealing the two isolated chambers for the in-ear and outer ear microphone; (C) Steps to assemble the earpiece by soldering the cable and speaker and then gluing the 3D-printed enclosure to create an ear-tight seal.

3.2.2 Web Dashboard and Mobile App. The existing OpenEarable web dashboard⁵ and mobile app⁶ were adapted to detect the 1.4 version of the device. If the respective version is detected, it is possible to configure the sampling rate of the inward- and outward-facing microphones via the web dashboard and app.

4 EVALUATION

The microphones used in OpenEarable 1.4 is an ultrasound capable microphone for active sensing applications. However, some emerging earable sensing applications use low frequency passive sounds from in the ear canal exploiting the occlusion effect [19]. Some examples include heart rate monitoring [5], step detection and interaction with facial gestures [11].

To show the suitability of the new earpiece for such applications an experiment into whether in-ear heart sounds and thus a measurement of heart rate is feasible was conducted from a single subject. Figure 3 shows how the device successfully captures the in-ear heart sounds and that the earpiece is occluding the ear canal by comparing the audio signal to the location of ECG R peaks from a simultaneously recorded signal taken from a Polar H10 device [1].

Additionally, the programmable gain was demonstrated on the task of capturing in-ear step sounds with prior work using such sounds for step counting [11] and user authentication [7]. This experiment is shown in Figure 4 where the correct choice of gain creates footstep sounds that are clearly resolved and not clipped.

5 POSSIBLE APPLICATIONS

Having direct access to earbud microphones via an open-source platform creates many possibilities for wearable computing research.

The in-ear mic has many sensing applications, while the outer microphone can be used for private speaking with a gesture over the mouth [21]. Both microphones have been used together for breathing monitoring during running [9], with the outer mic aiding in noise cancellation.

Applications include hearing aids and high sound pressure level warning systems. For instance, sound environment and level significantly impact mood [22] and mental health [18]. An external mic

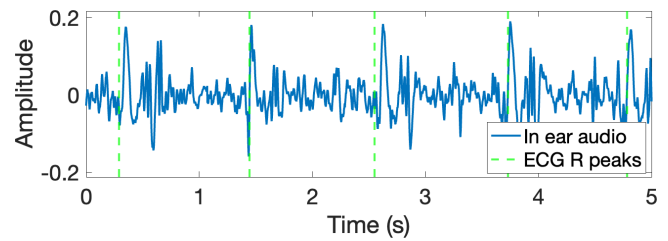


Figure 3: Waveform of in-ear heart sounds captured from the earpiece recorded with foam eartips (Etymotic ER1-14A) and maximum gain. Ground truth from a Polar H10 ECG strap annotated with green lines representing ECG R peaks.

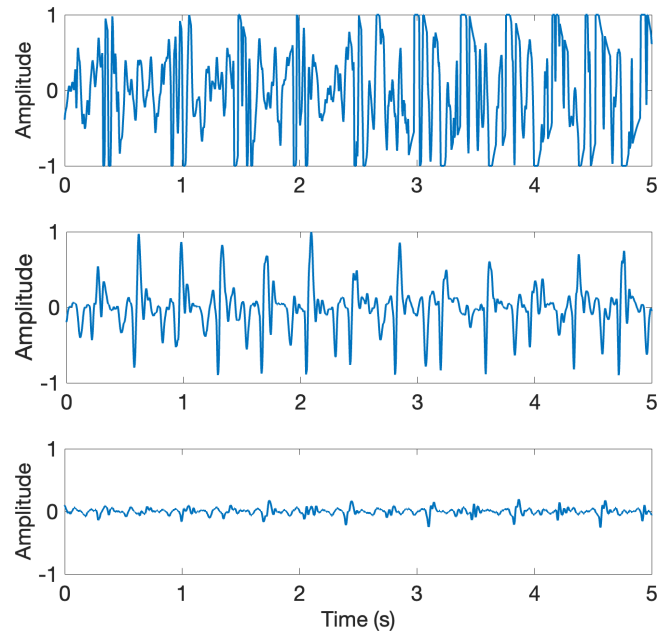


Figure 4: Examples of in-ear step sound recordings with gain changes, showing peaks for every step taken. (Top) maximum gain; (Middle) medium gain; (Bottom) minimum gain.

⁵<https://openearable.github.io/dashboard/>

⁶<https://github.com/OpenEarable/app>

can assess the soundscape and improve mood assessment frameworks like Mood Recorder [6]. Potential also exists in environmental sensing and epidemiological research. While air quality sensors have been explored [4], a wearable environmental noise measurement system is not yet available. Guidelines [3] requiring noise maps based on long-term measurements do not account for individual exposure, which is crucial for assessing health risks like cardiovascular issues [12].

Preliminary tests show that the 1.4 earpiece's ear canal pressure sensor enhances detection of phenomena like chewing activity [2] or ear rumbling [15].

Overall, we hope the improvements done to OpenEarable will help the community to pursue new research.

6 CONCLUSION

In this paper, we have presented an enhancement to OpenEarable through the introduction of a novel earpiece equipped with dual microphones. This new design addresses the previous limitation of OpenEarable by enabling the simultaneous capture of audio both inside the ear canal and from the surrounding environment. By leveraging the 8-pin connector, this earpiece facilitates novel auditory research and algorithm development. Our evaluation shows the capability to capture in-ear audio signals inline with recent research on in-ear heart sound and step detection. Using audio signals from inside and outside the ear will enable novel sensing applications and promises more robust audio-based sensing use cases. For example, external and internal noises measured by one or the other microphone could be filtered out.

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